

Question	Answer	Marks
4(a)	(micro)wave (from the transmitter) reflects (at metal/sheet)	B1
	The incident and reflected waves superpose	B1
	(resultant) amplitude is maximum at an antinode	B1
	(resultant) amplitude is minimum/zero at a node	B1
4(b)(i)	$\lambda = c / f$ $= 3 \times 10^8 / 6.3 \times 10^9$	C1
	= 0.048 m	A1
4(b)(ii)	distance PQ = $\frac{1}{4}\lambda$ $= 0.048 / 4$ $= 0.012 \text{ m}$	A1

Question	Answer	Marks
4(b)(iii)	(Distance QR is the) same (as PQ) <u>and</u> one of: • Distance (between maxima / minima) does not depend on intensity • distance depends <u>only</u> on wavelength • wavelength is unchanged / constant	B1

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